PROJECT DOCUMENT

Mitsue Project

Plan for the Maximum Introduction of Renewable Energy

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Plan for the Maximum Introduction of Renewable Energy

Clean English translation of key pages (Overview edition)

Source document: 御杖村再エネ導入最大化計画（概要版）, Mitsue Village (御杖村), Nara Prefecture · January 2025 (Reiwa 7) **Publisher:** Mitsue Village Policy Promotion Division (御杖村政策推進課)

> **About this translation.** This is a clean, human-readable English rendering of the key pages of the village's official plan, prepared for the Mitsue Project's outreach and grant work. It corrects the garbled machine translation in `20250130saienekeikakugaiyou_translated_eng-1.pdf`, in which the village name was rendered inconsistently as "Mitsu / Otsuke / Gose / Mikawa / Gozumura" — all of these are **御杖村 (Mitsue-mura)**. Page numbers refer to the source PDF. This is a faithful summary translation of the substantive content, not a certified line-by-line translation.

Cover (p.1)

Mitsue Village — Overview: Plan for the Maximum Introduction of Renewable Energy. January 2025 (Reiwa 7). Mitsue Village, Nara Prefecture.

Purpose, Position & Period (p.2)

Purpose of the plan. In recent years, severe natural disasters have increased worldwide due to climate change and extreme weather driven by global warming. The rise in temperatures shows no sign of stopping, and an era of "global boiling" is said to have begun. This plan aims to realize a decarbonized society by 2050 (Reiwa 32) by introducing renewable energy suited to the village's characteristics, undertaking village-wide global-warming countermeasures, and increasing forest absorption — showing a path toward carbon neutrality. At the same time, the village aims for a sustainable, decarbonized form of village management that enhances village vitality, residents' well-being, and secure living — making residents and future generations happy in a way unique to Mitsue.

Position of the plan. This plan is aligned with national and Nara Prefecture plans, and with the village's higher-level plans (the Mitsue Village Long-Term Comprehensive Plan and the First Mitsue Village Global Warming Countermeasures Implementation Plan). It is formulated as an integral part of the Global Warming Countermeasures Implementation Plan ("Regional Measures" edition), setting out a scenario for a carbon-neutral, sustainable society looking ahead to 2050.

Plan period. Reiwa 6 (2024) through Reiwa 32 (2050), with a six-year action window framed around the 2030 (Reiwa 12) countermeasures.

Basic Concepts for Carbon Neutrality (p.3)

1. **Reduce energy consumption** — thorough energy conservation by businesses and residents (efficient equipment, low-CO_2 transport, energy-saving devices).
2. **Transform energy use** — switch the type of energy used for heat or power to cut CO_2 (e.g., gasoline vehicles → EVs; kerosene heaters → wood-biomass stoves).

3. **Decarbonize energy** — introduce solar, wind, hydro, geothermal, and woody biomass that emit no CO_2 in generation.
4. **Sinks & offsets** — while advancing 1–3, enhance CO_2 absorption through proper forest conservation and management, leveraging the village's forest-rich character for long-term decarbonization.

Residents, businesses, and government can each start with initial actions; the goal is to popularize renewable energy that does not depend on fossil fuels, with the village's rich forests acting as a long-cycle CO_2 sink.

The Global-Warming Situation Around Mitsue (p.4)

Long-term records from observation sites near Mitsue show annual average temperature trending upward — **approximately +2.1°C over more than 42 years** — indicating warming is progressing. (Source: Japan Meteorological Agency "Meteorological Statistics," Ouda observation site.)

Village initiatives already under way. Solar power installed for nurseries and for housing for young single residents; storage batteries installed at nurseries; young-singles housing built to Nearly Zero Energy House (NearlyZEH) standard; and a wood-fired boiler installed at the tourist facility Mitsue Onsen "Himeshi no Yu" to reduce fossil-fuel use.

Emissions & Reduction Target (p.5)

Emissions. Mitsue Village emits about **9 thousand t-CO_2**. By sector, **transport is the largest at 46%** — higher than the national and prefectural averages — followed by residential 20%, industrial 18%, other-business 13%, and waste 3%.

Target. Compared with the 2013 (Heisei 25) baseline, the village aims for a **60% reduction by 2030 (Reiwa 12)** and to achieve **carbon neutrality by 2045 (Reiwa 27)** — surpassing the national 2050 target. Emissions fall from 13.2 (2013) → 9.4 (2021) → 5.3 (2030) thousand t-CO_2, with the remainder closed by energy saving, renewables, and forest absorption.

Three Key Initiatives (pp.6–8)

1. Local production and consumption of energy unique to Mitsue

Create and use renewable energy in harmony with the village's geography and natural conditions, so that local production and consumption become a value inherited into the future. Because energy generated within the village can be used during disasters and emergencies, build a distributed energy system effective in both peacetime and emergencies. **Key actions:** survey suitable sites and introduce small-scale hydropower using simple water-supply intake points; install solar PV on public facilities and roofs; **build a system to charge batteries and EVs from small hydro and solar, usable in both peacetime and emergencies; use V2H to consume EV-stored electricity in buildings.**

2. Use the village's rich forests from a decarbonization perspective

Maximize the use and enhancement of the village's forests as a new added value, with a village-wide focus on sustainable circular use. To increase CO_2 absorption, clarify forest ownership and boundaries and advance proper, planned forest maintenance; **if absorption is secured, explore introducing a J-Credit system** that lets forest carbon be bought and sold as added value. **Key actions:** create "Forest Lot Maps," promote cadastral (地籍) surveys, survey owners' intentions; support forest-management plans; develop forestry workers and a "selfharvesting forestry" style suited to Mitsue; in anticipation of future J-Credit generation and sales, consolidate forests into capable management bodies through multi-sector collaboration.

3. Increase value and pride as an environmentally conscious village

As the world moves toward carbon neutrality, the village's decarbonization efforts raise its value, so residents can take pride and act voluntarily. **Brand an environmentally gentle "eco rural living"** as a new appeal of the village to attract new residents and settlers, and to secure forestry successors; incorporate decarbonization into vacant-house use and housing-environment improvement; enhance learning opportunities; and make residents', businesses', and the community's eco-conscious actions and results visible.

Five Basic Policies & Measures (p.9)

1. **Promotion of energy-saving measures** — energy-saving actions; high-efficiency equipment; ZEB/ZEH building performance.
2. **Sustainable village building through local production for local consumption of renewable energy** — promote RE introduction; **form an autonomous, distributed energy society for disaster preparedness.**
3. **Promotion of low-carbon mobility** — promote EVs (EV/PHV/PHEV/FCV); low-carbon mobility.
4. **Formation of a circular society** — cut CO_2 by reducing waste; address solar-panel disposal.
5. **Decarbonization by all actors** — transition to decarbonized lifestyles/business; global-warming prevention measures; **CO_2 absorption by forests.**

Notable measures within these policies include: EV chargers at public facilities (town hall, roadside stations) **as a priority**; renewable energy from new technologies including **perovskite**; **solar carports**; PPA projects with no upfront cost; treating EVs/PHEVs as "mobile storage batteries"; off-grid capability to supply public buildings and clinics in emergencies; **VPP (Virtual Power Plant) using IoT (including blockchain)** to stabilize the grid; J-Credits via inter-municipal cooperation; and using thinning wood as fuel with the spread of wood stoves.

Selected Indicators & Targets (p.18)

Indicator	2024	2030	2045
HEMS (Home Energy Management System) adoption	0.5%	6.6%	25.0%
Home solar power use	4.1%	6.0%	14.0%
Resilient systems (renewables + storage + EV, etc.)	—	—	**One site within the village**
Hybrid / electric vehicle adoption	30.4%	58.2%	100%
Fuel-efficient ("eco") driving	64.1%	91.7%	100%
Reusable bag use / refusing plastic bags	71.0%	80.0%	100%
Households on a renewable-energy electricity menu	8.8%	30.0%	100%

Indicator	2024	2030	2045
Forest absorption via forest maintenance	**0.3 kt/yr**	**1.8 kt/yr**	**2.7 kt/yr**

(Selected from the full indicators table; values shown as current / 2030 target / 2045 target.)

Promotion System (p.19)

Achieving the targets requires all actors — **Village government, Businesses, and Residents/Community** — to cooperate under an "ALL Mitsue" approach. A (tentative) Mitsue Village Decarbonization Promotion Council of experts, regional representatives, and businesses meets regularly to review and revise the plan. A Promotion Headquarters is led by the village chief (head) and deputy chief; Promotion Officers are assigned in each department; the Secretariat sits in the Resident Life Division and manages overall progress.

Grant Lineage & Contact (p.20)

> Created under the **Ministry of the Environment** grant project for FY2023 (supplementary budget): subsidies for CO_2-emission-reduction measures — *"Regional decarbonization to enable large-scale introduction of renewable energy: planning-support project"* — funded by a grant to the **Association for Regional Circular and Ecological System (地域循環共生圏).**

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(The plan also includes a glossary defining eco-driving, greenhouse gases, carbon offsetting, carbon neutrality, renewable energy, J-Credits, self-sufficient/distributed energy society, EV, V2H, and wood-based biomass.)

Clean translation prepared by the Mitsue Project — June 7, 2026. For the project's own analysis of how this plan aligns with the project, see `mitsue_village_re_plan_alignment.md`.